# Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

## Interaction

| | |
|---|---|
| **Meaning in my words**<br>How would you explain interaction to a friend? | **How can you tell?**<br>In “the rope pulls the crate,” which two objects are interacting? |
| **Example or sketch**<br>What interaction holds up a book resting on a table? | **Non-example and why**<br>Someone names “moving right” as the object pushing a cart. What would you ask? |

## Contact force

| | |
|---|---|
| **Meaning in my words**<br>How would you explain contact force to a friend? | **How can you tell?**<br>Which detail tells you the table’s push on a book is a contact force? |
| **Example or sketch**<br>Draw the force a compressed spring exerts on a cart touching its right end. | **Non-example and why**<br>Earth pulls on a falling ball after it leaves your hand. Is that a contact force? |

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## Non-contact force

| | |
|---|---|
| **Meaning in my words**<br>How would you explain non-contact force to a friend? | **How can you tell?**<br>A magnet tugs a paper clip before they touch. How can you tell contact is unnecessary? |
| **Example or sketch**<br>Which downward arrow belongs on a ball after you let go? | **Non-example and why**<br>A hand holds up a book. Which force disappears when the hand moves away? |

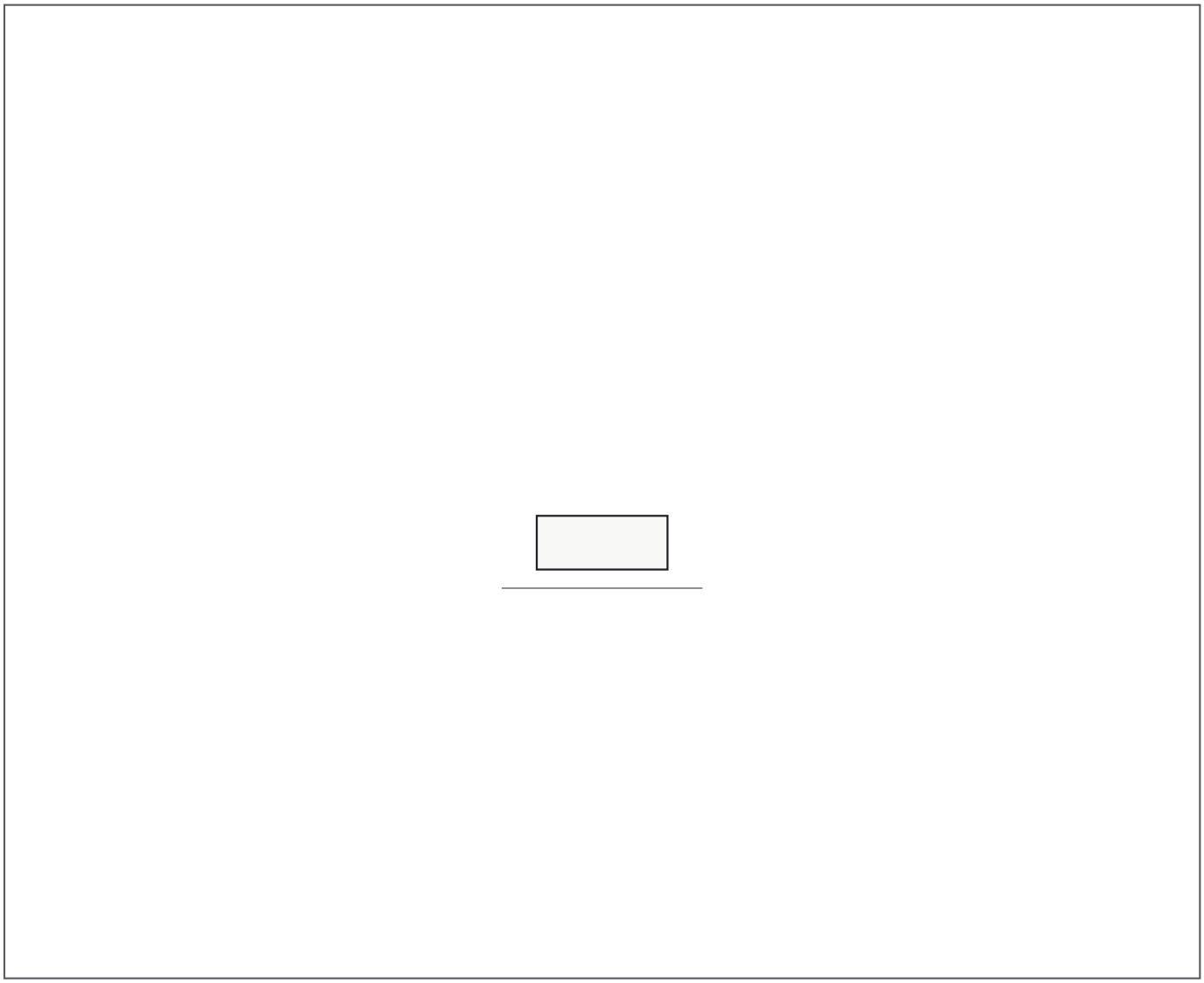
## Mass vs. weight

| | |
|---|---|
| **Meaning in my words**<br>How would you explain the difference between mass and weight? | **How can you tell?**<br>What changes when you take the same cart to the Moon: its mass or its weight? |
| **Example or sketch**<br>On Earth, use 10 N/kg. What are the mass and weight of a 2 kg cart? | **Non-example and why**<br>A label says “mass = 20 N.” What does the unit tell you? |

# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

## Draw Earth and a crate. Put weight on the crate; label source, receiver, direction, and N.



My marks mean

# On assignment: Interaction

SECTION E / BACK / FINAL MEETING ONLY Name _____

**An XJ animation shows gravity from Earth and a forward arrow labeled “because it moves.” The graphics desk needs approved force labels.**

## PICK A JOB / DELIVER ONE PRODUCT

☐ **Simulation engineer**

Prepare the force layer for XJ. Name a source and receiver for each arrow you keep. Put motion on a separate layer.

☐ **Science graphics editor**

Sketch the graphic viewers will see. Have your producer challenge the source of each arrow before you approve its label.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

## INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.